

3(a)	any two from: 1 (infection / treatment with) <u>GM</u> <i>M. pingshaense</i>, reduces / AW, the (percentage of mosquitoes that) produce eggs ; 2 error bars for control and <u>GM</u> (<i>M. pingshaense</i>) <u>do not overlap</u>, therefore (mean percentage of mosquitoes that produce eggs) is <u>significantly different</u> ; 3 <i>conclusion for non-GM M. pingshaense</i>: error bars for <u>non-GM</u> (<i>M. pingshaense</i>) and control overlap, therefore (mean percentage of mosquitoes that produce eggs) (may be) <u>not significantly different</u> or error bars for <u>non-GM</u> (<i>M. pingshaense</i>) and <u>GM</u> (<i>M. pingshaense</i>) <u>overlap</u>, therefore (mean percentage of mosquitoes that produce eggs) (may be) <u>not significantly different</u> or large / AW, error bar for <u>non-GM</u> (<i>M. pingshaense</i>) shows that (calculated / sample) <u>mean</u> is (likely to be) further from the, true / actual, <u>mean</u> or <i>idea that</i> error bar, for <u>non-GM</u> (<i>M. pingshaense</i>) are, large / AW, and the <u>mean</u> is less reliable ;	2 allow fungus as AW for <i>M. pingshaense</i> allow <i>A. coluzzii</i> / <i>Anopheles</i> as AW for mosquitoes allow uninfected <i>A. coluzzii</i> as AW for control 1 ignore ref. to <u>non-GM</u> <i>M. pingshaense</i> ignore kills mosquitoes 2 allow (95%) confidence intervals / CI / standard error bars as AW for error bars ignore standard deviation / SE / standard error unqualified allow significant, decrease / AW (implies a difference) allow may be significantly different ignore 'results are significant' 3 allow (95%) confidence intervals / CI / standard error bars as AW for error bars ignore standard deviation / SE / standard error unqualified allow is not significantly different ignore no significant decrease ignore insignificant ignore data / results / values as AW for mean ignore accurate / valid / precise
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Question	Answer	Marks	Guidance																														
3(b)(i)	<i>idea of</i> placing a, net / glasshouse / AW, over the experimental area ;	1	allow web / mesh ignore (use) black fabric / curtain ignore close doors / windows of hut ignore insect repellent R insecticide																														
3(b)(ii)	1. Table 3.2 completed and correct values of $(O-E)^2 / E$; 2. $(\chi^2 =)$ 2390 when rounded to 3 significant figures ; 3. $(\chi^2 =)$ 2390 ; must be to <u>3 significant figures</u> <table><tr><th>repeat</th><th><i>O</i></th><th><i>E</i></th><th><i>O - E</i></th><th>$(O - E)^2$</th><th>$\frac{(O - E)^2}{E}$</th></tr><tr><td>1</td><td>13</td><td>1396</td><td>-1383</td><td>1912689</td><td>1370.1</td></tr><tr><td>2</td><td>61</td><td>731</td><td>-670</td><td>448900</td><td>614.1</td></tr><tr><td>3</td><td>12</td><td>429</td><td>-417</td><td>173889</td><td>405.3</td></tr><tr><td>total:</td><td></td><td></td><td></td><td></td><td>2390</td></tr></table>	repeat	<i>O</i>	<i>E</i>	<i>O - E</i>	$(O - E)^2$	$\frac{(O - E)^2}{E}$	1	13	1396	-1383	1912689	1370.1	2	61	731	-670	448900	614.1	3	12	429	-417	173889	405.3	total:					2390	3	1. A as fractions / any decimal places 2. correct value of χ^2 to 3 or more sig figs use calculation guidance – give mp2 if their value of χ^2 rounds to 2390 (e.g. 2389.5, 2389.4, 2389) 2. and 3. allow ecf to max 2 from incorrect $(O-E)^2 / E$
repeat	<i>O</i>	<i>E</i>	<i>O - E</i>	$(O - E)^2$	$\frac{(O - E)^2}{E}$																												
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3(b)(iii)	<i>critical value:</i> 5.991 ($p = 0.05$ and $df = 2$) and <i>explanation:</i> yes / student was correct / reject null hypothesis / difference is significant and calculated value / $\chi^2 / 2390$, is greater than, critical value / 5.991 ; ora	1	allow critical value from table (e.g. underlined / circled) allow 9.210 ($p = 0.01$ and $df = 2$) allow ecf if incorrect χ^2 must give correct statistics for ecf: e.g. calculated value, is less than 5.991 and null hypothesis is accepted / student is incorrect
3(c)	<i>supports the conclusion:</i> <i>using Table 3.1:</i> 1 (total) number of, <i>A. coluzzii</i> / mosquitoes is, lower / AW, with (GM) <i>M. pingshaense</i> ; ora <i>using calculated value of χ^2:</i> 2 there is a <u>significant difference</u> ;	4	allow fungus as AW for (GM) <i>M. pingshaense</i> allow <i>A. coluzzii</i> / <i>Anopheles</i> as AW for mosquitoes 1 ignore ref. to egg production / reproduction 2 ignore reject null hypothesis A significant decrease R correlation / relationship A less than, 5% / 0.05 / 1% / 0.01, probability that, difference is due to chance A ora greater than, 95% / 0.95 / 99% / 0.99, probability that difference is not due to chance ecf from incorrect χ^2 A ecf there is no <u>significant difference</u> (if calculated value of χ^2, is less than 5.991)

Question	Answer	Marks	Guidance
3(c)	<i>further information needed (max 3):</i>		ignore repeat investigation unqualified ignore larger sample size ignore do a control ignore <i>ref. to</i> statistical test ignore do investigation in other areas / countries ignore <i>ref. to</i> cost
	3 carry out investigation in other years ;		3 ignore longer time unqualified ignore carry out investigation in other seasons
	4 carry out investigation with other species of (<i>Anopheles</i>) mosquito ;		4 allow other types / varieties / AW, of (<i>Anopheles</i>) mosquito
	5 <i>idea of</i> checking if, (GM) <i>M. pingshaense</i> could be, harmful / toxic / AW, to humans ;		5 and 6 ignore (GM) <i>M. pingshaense</i> could be harmful unqualified
	6 <i>idea of</i> checking if (GM) <i>M. pingshaense</i> could be harmful / toxic / AW, to, livestock / ecosystems / biodiversity / (native) species / AW ;		
	7 <i>idea of</i> checking if, <i>A. coluzzii</i> / mosquitoes, become resistant to GM <i>M. pingshaense</i> ;		
	8 <i>idea of</i> checking how long (GM) <i>M. pingshaense</i> survives (on the black cotton fabric) ;		8 A checking how often (GM) <i>M. pingshaense</i> should be applied (to the black cotton fabric)